

Chapter 1

Mean Value and Intermediate Value Theorems

1.1 Extrema and Critical Points

Definition 1.1 (Global and Local Extrema). Let f be defined on a domain D .

1. f has an absolute (or global) maximum at $c \in D$ if $f(c) \geq f(x)$ for all $x \in D$.
2. f has a local maximum at c if there exists an open interval I containing c such that $f(c) \geq f(x)$ for all $x \in I \cap D$.

Minimums are defined analogously.

Definition 1.2 (Critical Point). A point c in the domain of a differentiable function f is called a critical point if either $f'(c) = 0$.

Theorem 1.3 (Fermat's Theorem). *Let f be a function which is defined and differentiable on the open interval (a, b) . If $c \in (a, b)$ is a maximum (or minimum) for the function, then $f'(c) = 0$.*

Example 1.4. Let $f(x) = x^3 - 3x$. We find the critical points by solving $f'(x) = 0$. We have $f'(x) = 3x^2 - 3 = 3(x - 1)(x + 1)$. Thus, $x = 1$ and $x = -1$ are critical points. Evaluating the function, $f(1) = -2$ (a local minimum) and $f(-1) = 2$ (a local maximum).

1.2 The Mean Value Theorem

Theorem 1.5 (Extreme Value Theorem). *If a function f is continuous on a closed and bounded interval $[a, b]$, then f attains an absolute maximum value $f(c)$ and an absolute minimum value $f(d)$ at some numbers $c, d \in [a, b]$.*

Theorem 1.6 (Rolle's Theorem). *Let f be continuous on the closed interval $[a, b]$ and differentiable on the open interval (a, b) . If $f(a) = f(b)$, then there exists $c \in (a, b)$ such that $f'(c) = 0$.*

Theorem 1.7 (Mean Value Theorem). *Let f be continuous on the closed interval $[a, b]$ and differentiable on the open interval (a, b) . Then there exists $c \in (a, b)$ such that*

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Example 1.8. Consider $f(x) = x^2$ on the interval $[1, 3]$. The function is continuous on $[1, 3]$ and differentiable on $(1, 3)$. The average rate of change is $\frac{f(3) - f(1)}{3 - 1} = \frac{9 - 1}{2} = 4$. By the Mean Value Theorem, there exists $c \in (1, 3)$ such that $f'(c) = 4$. Since $f'(x) = 2x$, solving $2c = 4$ yields $c = 2$, which lies in $(1, 3)$.

Theorem 1.9 (Monotonicity and Derivatives). *Let f be differentiable on an interval I .*

1. *If $f'(x) > 0$ for all $x \in I$, then f is strictly increasing on I .*
2. *If $f'(x) < 0$ for all $x \in I$, then f is strictly decreasing on I .*
3. *If $f'(x) = 0$ for all $x \in I$, then f is constant on I .*

1.3 Intermediate Value Theorem

Theorem 1.10 (Intermediate Value Theorem). *Let f be a continuous function on the closed interval $[a, b]$. If y is any number strictly between $f(a)$ and $f(b)$, then there exists $c \in (a, b)$ such that $f(c) = y$.*

Example 1.11. We show that the equation $x^3 - x - 1 = 0$ has a root in the interval $[1, 2]$. Let $f(x) = x^3 - x - 1$. Since f is a polynomial, it is continuous everywhere. We evaluate the endpoints: $f(1) = 1 - 1 - 1 = -1$ and $f(2) = 8 - 2 - 1 = 5$. Since $f(1) < 0 < f(2)$, the Intermediate Value Theorem guarantees the existence of a $c \in (1, 2)$ such that $f(c) = 0$.